

Kevin González Ortega

Data Scientist | Psychometrics and Educational Measurement | Machine Learning and AI Evaluation

Bloomington, IN, USA | kevingpanama@gmail.com | linkedin.com/in/kevingpanama

SUMMARY

Data scientist and Systems and Computing Engineer specializing in psychometrics and educational measurement. Fulbright Scholar pursuing an M.S.Ed. in Qualitative and Quantitative Research Methodology at Indiana University Bloomington. Ten years of experience building data and learning systems at Fundación Ayudinga, where I lead technology and educational research, direct an R&D portfolio of 10+ projects in machine learning, knowledge management and gamification for education, and designed the data governance strategy. I combine software engineering, statistical modeling and measurement to build and evaluate models and assessments that are valid, reliable and fair.

EDUCATION

Indiana University Bloomington, Bloomington, IN, USA

Master of Science in Education (M.S.Ed.), Qualitative and Quantitative Research Methodology | Aug 2025 – May 2027 (expected)

- Fulbright Scholar. Emphasis in measurement, multivariate statistics and machine learning for the social sciences.
- Completed coursework: Intermediate Statistics Applied to Education; Multivariate Analysis in Educational Research; Strategies for Educational Inquiry; Qualitative Inquiry in Education; Computer Laboratory and Consultation for Educational Inquiry; Network Science; Education Policy and Reform.
- Current coursework (Fall 2026): Educational Assessment and Psychological Measurement; Machine Learning Methods for Education and Social Science; Applied Missing Data Analysis.
- Planned (Spring 2027): Longitudinal Data Analysis; Master's Thesis in Education.

University of North Texas, Intensive English Language Institute, Denton, TX, USA

Fulbright Pre-Academic English Program | Jan 2025 – Jul 2025

Universidad Tecnológica de Panamá, Panama City, Panama

Bachelor of Engineering, Systems and Computing Engineering | Mar 2016 – Dec 2023

FELLOWSHIPS AND HONORS

Fulbright Faculty Development Program Scholar, U.S. Department of State | Jul 2025

Awarded to pursue graduate study in research methodology at Indiana University Bloomington, preceded by an intensive English program at the University of North Texas.

PROFESSIONAL EXPERIENCE

Fundación Ayudinga, Panama

Director of Technology and Educational Research | Jun 2019 – Present

- Lead technology and educational research at a nonprofit e-learning foundation.
- Direct an R&D portfolio of 10+ projects in machine learning, knowledge management and gamification for education.
- Lead research on factors that influence learning, analyzing experimental data to build and evaluate artificial intelligence models.
- Designed the data governance strategy (data architecture, data quality and business intelligence) and the platform's cloud infrastructure, including the development and maintenance of cloud services and web servers.
- Plan web and desktop application projects and lead systems analysis and design.
- Advise student research, including a gamified, adaptive tool for learning fractions that accounts for working memory and math anxiety, presented at Universidad Tecnológica de Panamá's Jornada de Iniciación Científica (JIC).

Director of Finance and Treasurer of the Board | Feb 2020 – Present

- Built the foundation's finance function from scratch and consolidated its accounting information.
- Automated financial reporting for decision-making; prepare and manage the annual budget.

Backend Lead | Aug 2017 – Jun 2019

- Led the backend team of the foundation's e-learning platform.

Junior Software Developer | Feb 2016 – Jul 2017

- Developed features for the foundation's e-learning platform.

TEACHING EXPERIENCE

Universidad Interamericana de Panamá, Panama City, Panama

Adjunct Professor (Part-time) | Aug 2023 – Dec 2024

- Taught undergraduate Discrete Structures and Precalculus.

UTP-FISC Tech Academy, Universidad Tecnológica de Panamá, Panama City, Panama

Volunteer Instructor | Feb 2018 – Mar 2020

- Designed, planned and taught Python for Programmers (2020), Introduction to Python (2019) and NoSQL Databases (2018).

RESEARCH AND PROJECTS

Gamified, Adaptive Learning Tool for Fractions, Fundación Ayudinga

Research Advisor

- Co-advised, with Johel Batista, a student team designing an educational tool for learning fractions that combines gamification, cognitive training, spaced repetition and adaptive learning.
- The design was informed by working memory and math anxiety, two factors that affect performance in mathematics. Presented at the Jornada de Iniciación Científica (JIC), Universidad Tecnológica de Panamá.

Cambio en 10 | Mar 2018 – Aug 2018

International Laboratory for Citizen Advocacy

- Project to reduce school dropout by developing life skills, with emotional intelligence as a cross-cutting factor of comprehensive education.

PUBLICATIONS

Opinion columns on education policy, *La Prensa* (Panama), in Spanish

- González Ortega, K. (2019, January 15). Queremos espacio.
- González Ortega, K. (2018, December 18). Nuestra voz fue escuchada.
- González Ortega, K. (2018, October 23). En el filo de la navaja.
- González Ortega, K. (2018, September 25). Oportunidades para todos.
- González Ortega, K. (2018, July 17). Fui desertor.
- González Ortega, K. (2018, June 12). El mundial que hay que jugar.
- González Ortega, K. (2018, April 3). Hora de alzar la voz.
- González Ortega, K. (2018, February 13). El entierro de la sardina.
- González Ortega, K. (2017, October 24). Medir para saber, entender y mejorar [Measuring to know, understand and improve].
- González Ortega, K. (2017, August 22). Para disminuir la desigualdad educativa [To reduce educational inequality].

LEADERSHIP AND SERVICE

Jóvenes Unidos por la Educación, Panama

Selection and Recruitment Coordinator | Nov 2018 – Aug 2020

- Built the applicant selection and recruitment system for the Latin American Citizen Action Laboratory and its interactive virtual platform.

Operations Team Member | Nov 2016 – Nov 2018

- Co-organized #HablemosDeEducación19 (Aug 2018 – May 2019), a citizen advocacy campaign that proposed 19 education policy measures for Panama's 2019 general elections and set out to measure their execution by the next government.
- Co-organized the first and second editions of "The Role of Youth in the Future of the Panama Canal," held with the Panama Canal Authority, in which 400+ young people presented their views on the Canal's challenges and future.

Laboratorio Latinoamericano de Acción Ciudadana, Panama

Mentor, AC7IVATE! | Jun 2019 – Dec 2021

- Mentored a socio-emotional learning program for students aged 11–12 built around six challenges: self-esteem, recognizing and accepting emotions, assertive communication, teamwork, conflict resolution and critical thinking.

CERTIFICATIONS

- International Diploma in Data Protection Regulation in Panama, Universidad de Panamá, Apr 2025 (Credential ID VIEX-62932-D)
- Big Data Integration and Processing, University of California San Diego (Coursera), Nov 2019 (Credential ID G75XSE3CL7A2)
- Introduction to Big Data; Big Data Modeling and Management Systems; Machine Learning with Big Data, University of California San Diego (Coursera)
- The Data Scientist's Toolbox; R Programming, Johns Hopkins University (Coursera)

SKILLS

Programming and Data: Python, R, SQL, NoSQL, JavaScript, C++ , MATLAB, .NET, HTML, Dataiku DSS, Microsoft Excel

Statistics and Measurement: Multivariate Statistics, Educational Measurement, Psychometrics, Statistical Modeling, Missing Data Analysis, Quantitative Research, Qualitative Research, Research Design

Machine Learning and AI: Machine Learning, Natural Language Processing (NLP), Large Language Models (LLM), Knowledge Graphs, Network Science

Data Management: Data Governance, Data Quality, Data Privacy Regulation, Business Intelligence, Big Data, Cloud Computing, Geographic Information Systems (GIS)

Education: University Teaching, Educational Technology, Learning Analytics

PROFESSIONAL MEMBERSHIPS

IEEE, Member (Oct 2025 – Present) | Association for Computing Machinery (ACM), Member (Mar 2019 – Present) | Project Management Institute (PMI), Member (Jan 2024 – Present)

LANGUAGES

Spanish (native) | English (professional; graduate study in English at Indiana University Bloomington)